

App Dev Project Report

1. Student Details

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About Me: I am a student at IITM, love to craft solutions for problem society face.

2. Project Details

Project Title: Niyukt

Problem Statement:

To make a placement portal application for college students and admins to manage their placement activity and drives, removing the problem of big excel based management.

Approach:

Made a simple application which could be used by company, student and college admins to manage the placement drives in their college, featuring good data base design with reliable logic which could sustain load for a basic college infrastructure hosting.

3. AI/LLM Declaration

No use of AI/LLM, Every part is hard crafted and made with joy to learn App dev.

4. Technologies and Frameworks Used

Technology / Library	Purpose
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Flask	Core backend web framework
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Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Flask-session	Very simple authentication system used
SQLite	Lightweight local database for storing user data
Passlib & argon2	Used for password hashing
matplotlib	Server based chart render service
numpy	Basic computational use
Browser API	Used JS to a good extend for part rendering in dashboard

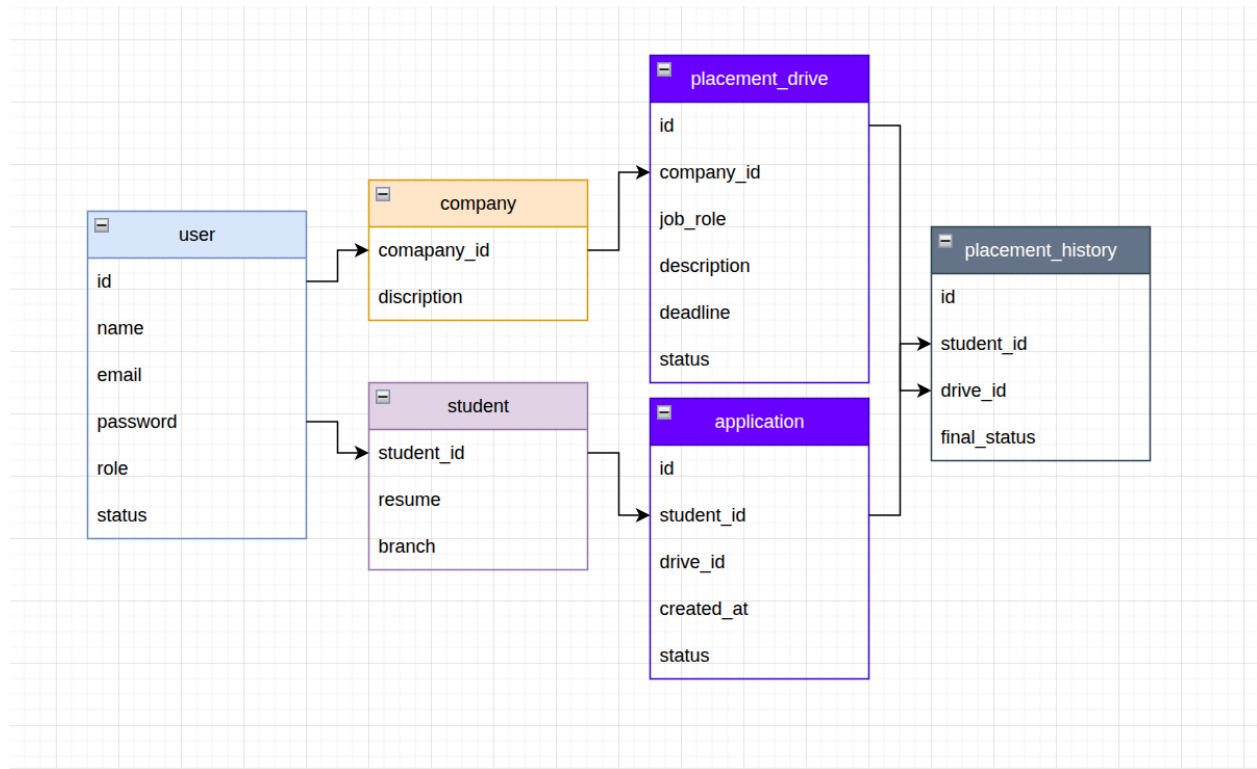
5. Database Schema / ER Diagram

Tables:

1. **user** — stores user profile details (id, name, email, password)
2. **placement_drive** — Drive created by company gets through admin validation for listing
3. **application**— Made by student to apply for a company job position

Relationships:

- One-to-Many → **User** → **Application/Drive/placement_history**
- One-to-One → **User** → **company, student**
- One-to-One → **Drive** → **student**
- One-to-Many → **company** → **placement_drive**
- One-to-One → **student** → **placement_history**



6. API Resource Endpoints

Endpoint	Method	Description
<code>/login</code>	POST	Authenticate user and simple session
<code>/register/company</code>	GET, POST	Register Company and wait for approval
<code>/register/student</code>	POST, GET	Student registration endpoint
<code>/admin</code>	GET	Admin Dashboard
<code>/admin/users</code>	GET	Fetch all users of application
<code>/admin/alter-status</code>	POST	Activation and deactivation endpoint for user
<code>/admin/requests</code>	GET	All drive requests for admin for approval
<code>/admin/alter-drive-status</code>	POST	Drive status alteration endpoint for admin

<code>/graphs</code>	GET	Loading Basic graphical summary for the application
<code>/company</code>	GET	Company dashboard page
<code>/company/application/<int:drive_id></code>	GET	Applications for given drives listing
<code>/company/alter-application</code>	POST	Alter application of students
<code>/company/create-drive</code>	POST, GET	Create placement drive and wait for approval via admin
<code>/student</code>	GET	Student dashboard all approved drives listing for application
<code>/student/apply/<int:drive_id></code>	POST	Apply into given listed drive
<code>/student/applications</code>	GET	All applications made by user

7. Architecture and Features (optional)

Architecture Overview:

- **app.py** – main Flask application entry point
- **/models** – database models using self made SQL engine
- **/api** – Flask Blueprints for all of the pages of application
- **/templates** – Jinja2 HTML templates and base templates
- **/static** – CSS, JS, and chart visualization files

Implemented Features:

User registration and login
Student apply option
Visualization dashboard using matplotlib lib
Summary analytics with date filtering

Responsive UI built with Bootstrap
Secure password hashing and user session handling

Additional Features for future

- Ai Mock interview portal for student to practice
 - Modern Learning portal for student to groww
 - Startup funding schema for students future leading
 - Launching this as a startup to boost indian students as whole 😊
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8. Video Presentation

Drive Link:

<https://drive.google.com/file/d/your-video-link-here>

(Accessible to all with "View" permission.)